Venkata Lakshmi Parimala Pasupuleti

Foundational AI Engineer · Tool-Calling Agents · Model Orchestration · RAG · MCP Integrations · Agentic Workflows · Regulatory Tech AI · Python · AWS

venkatalakshmiparimalap@gmail.com | +1-703-261-5410 | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | AgentProbe (open-source) | Virginia · Open to NYC

PROFESSIONAL SUMMARY

I build production AI systems end-to-end — **tool-calling agents, model orchestration, RAG pipelines** — for high-stakes automated decisions in regulated domains. 9+ years engineering experience (7+ in production); 2+ years building and shipping LLM systems. Currently at Intuit, owning the full **agentic pipeline** integrating Anthropic Claude into financial apps: **MCP-style tool integrations**, RAG grounding on customer financial records, LLM-as-judge eval CI (caught 3 prompt regressions pre-release, zero hallucination incidents post-deploy), observability via OpenTelemetry. Two open-source agentic systems: **AgentProbe** — multi-provider **model orchestration** across 5 LLM providers with LLM Router + **tool-calling agent framework** + 15-endpoint FastAPI backend; **Leasel** — **regulatory compliance automation** with 8-agent LangGraph StateGraph, **agentic workflows** for complex decision pipelines, hybrid RAG, 93.6% precision across 1,240-case eval harness. PostgreSQL (7 yrs), Redis, AWS in production at 100K+ MAU scale. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). — github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

Tool-Calling Agents | Model Orchestration | RAG Pipelines | MCP Integrations | Agentic Workflows | LLM Production Systems | Regulatory Tech AI | LLM Evaluation & Evals CI | Multi-Agent Architectures | Vector Databases | Automation Orchestration | Production AI Observability

EXPERIENCE

Intuit

Jun 2026 – Present · Virginia, USA

Software Engineer — LLM Production Systems · Financial Services AI · MCP-Style Integrations · Python · AWS

- Own **LLM production systems** end-to-end at Intuit: integrate Anthropic Claude into financial apps via **MCP-style tool integrations** and structured-output schemas, with RAG grounding on customer financial records for full citation traceability; every response traceable to source data.
- Build **agentic workflows** for financial data summarization and natural-language queries: **tool-calling** pipelines route user intent to financial APIs, Claude reasons over retrieved context, deterministic validators gate outputs before surfacing to users.
- Own **eval CI for LLM outputs**: LLM-as-judge scoring with golden datasets and regression detection; caught 3 prompt regressions before release; zero hallucination incidents post-deployment; prompt A/B testing + model comparison loops integrated into dev cycle.
- Build guardrails: input validation, sensitive-data redaction, fallback paths on model failure; containerize on Docker, ship via GitHub Actions to **AWS**; observability with OpenTelemetry, Prometheus, Grafana for full production AI visibility.

Entrata (Apartment & Affordable Housing SaaS)

Oct 2020 – Jul 2024 · 100K+ MAU, 1M+ property records

Senior Software Engineer — Python · Java · AWS · PostgreSQL · Redis · Distributed Systems

- Shipped 20+ production features end-to-end (Python, Java/Spring Boot, PHP, **PostgreSQL**, React/TypeScript) across 120+ Agile sprints; reviewed 300+ PRs; mentored junior engineers.
- Built **automation workflows** in Python/Bash eliminating 80% of manual operational work; reduced 200+ **PostgreSQL** queries' runtime 30% via execution plan analysis and indexing; **Redis**-backed async processing with idempotency guarantees for bulk tenant record updates.
- Owned HUD income-limit ETL: federal source rows → 180K regulatory domain records with validation, rollback checks, compliance enforcement; 7 annual cycles, zero violations — **regulatory tech** compliance at production scale.
- **AWS S3** PHI migration with Libsodium encryption (100% PHI compliance, zero security incidents); Docker microservices redesign cutting execution time 50%; **RabbitMQ** async processing for high-volume tenant workflows.

Entrata India (Db Xento Systems)

Sep 2018 – Oct 2020 · Pune, India

Junior Software Engineer — Python · PHP · Java · PostgreSQL · AWS

- Built property management modules and compliance data pipelines processing 1.6M HUD/Tax Credit records; led AWS S3 cloud migration from legacy server storage; created 56 state-specific housing certification documents.

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

AI/ML Research · Fine-Tuning · PyTorch · HuggingFace · Statistical Modeling

- Fine-tuned classification model (HuggingFace Transformers + PyTorch): full training loop (data prep → fine-tuning → eval → deploy on Quart), presented to GMU faculty panel. Built predictive models ($R^2 = 0.977$); applied Isolation Forest anomaly detection. Mentored 600+ students.

PORTFOLIO PROJECTS

AgentProbe — Multi-Provider Model Orchestration & Tool-Calling Agent Framework

Model Orchestration · Tool-Calling Agents · MCP-Ready

2025 · Python, FastAPI (15 endpoints), LangGraph (3 agents), ChromaDB, Pinecone, Anthropic, OpenAI, Azure OpenAI, Groq, Ollama · github.com/venkatalakshmiparimala/AgentProbe

- **Multi-provider model orchestration** across 5 LLM providers (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama): LLM Router classifies task intent and dynamically routes to optimal provider based on latency, cost, and capability signals — **tool-calling agent framework** with inter-agent state handoffs via LangGraph StateGraph.
- **5-metric LLM-as-judge eval framework** (factual accuracy, hallucination rate, retrieval quality, completeness, compliance); CI-integrated regression detection blocks deploys when quality drops; RAG pipeline evaluation with **vector database** (ChromaDB + Pinecone); 15-endpoint FastAPI backend; used to evaluate Leasel across 1,240 test cases.

Leasel — Regulatory Compliance Automation with Agentic Workflows

Regulatory Tech AI · Agentic Workflows · RAG Pipeline

2026 · Python, LangGraph (8-agent StateGraph), FastAPI, hybrid RAG (TF-IDF + semantic reranking), SQLite, Anthropic/OpenAI APIs · github.com/venkatalakshmiparimala/Lease_complaince_rag

- **Regulatory compliance automation** via 8-agent LangGraph StateGraph: **agentic workflows for complex decision pipelines** — triage → retrieval → Config/History/StructuredData parallel agents → reasoning → deterministic validator → confidence-gated HITL escalation; correctly traces false-ineligibility ticket to misconfigured setting + release tag.
- **RAG pipeline** with hybrid TF-IDF + semantic reranking + section-aware chunking: 100% citation correctness, 0% unsupported-claim rate; **automation workflow** enforces regulated calls via deterministic logic (LLM reasons, validator decides); 93.6% precision, 87.9% F1 across 1,240-case eval harness — directly maps to tax/regulatory automation domain.

AI on FHIR — Clinical NLP & LLM API Integration

LLM Integration · HIPAA Guardrails

2025 · Python, Flask, React, TypeScript, Docker, BERT, spaCy, LLMs, FHIR REST APIs · github.com/venkatalakshmiparimala/AI_on_FHIR

- NLP pipeline translating English/Spanish clinical questions into FHIR API calls via LLM **tool-calling** and embeddings; HIPAA guardrails with RBAC for data isolation; ~60% reduction in manual clinical lookup time; containerized with Docker; React + **TypeScript** frontend.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

Agentic AI & LLMs: Tool-Calling Agents, Model Orchestration, Agentic Workflows, RAG Pipelines, MCP Integrations, LLM Production Systems, Multi-Agent Architectures, LangGraph StateGraph, LangChain, LLM-as-Judge, Eval CI/CD, LLM Evaluation Frameworks, HITL Escalation, Guardrails, Hallucination Detection, Structured Outputs, Prompt Engineering, Anthropic Claude, OpenAI, Azure OpenAI, Groq, Ollama, n8n-style Orchestration, Automation Workflows

Vector Databases & Retrieval: Pinecone, ChromaDB, Weaviate, Elasticsearch, HNSW, Semantic Search, Hybrid Retrieval, TF-IDF + Reranking, Embeddings, BERT, spaCy, HuggingFace Transformers, PyTorch

Engineering & Backend: Python (9+ yrs), FastAPI, Flask, Django, Java, Spring Boot, TypeScript, JavaScript, Node.js, React, PHP, SQL, PL/SQL, Bash, PostgreSQL, MySQL, Oracle, Redis, MongoDB, SQLite, Snowflake, ETL Pipelines, Data Modeling, Query Optimization, RabbitMQ, Kafka, Celery

Cloud, DevOps & Observability: AWS (S3, EC2, RDS, Lambda, Glue, SQS, API Gateway, IAM), Azure (OpenAI, Data Factory, Databricks, AKS), GCP (Vertex AI), Docker, Kubernetes, Terraform, GitHub Actions CI/CD, OpenTelemetry, Prometheus, Grafana